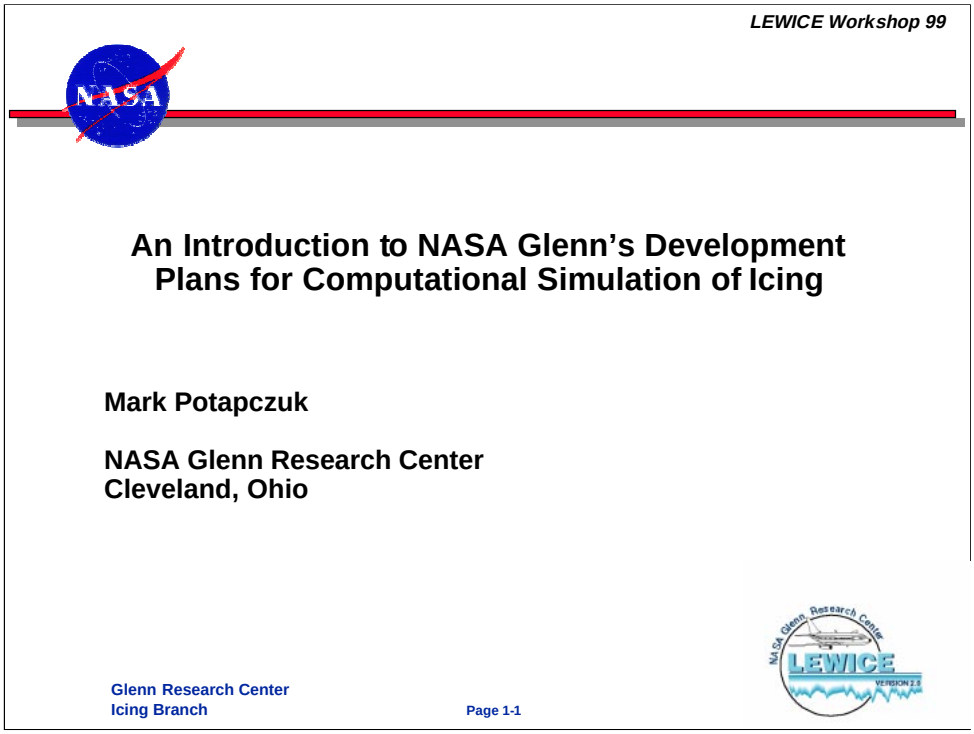


Welcome to the NASA Glenn Research Center LEWICE Workshop.

This workshop is being conducted to introduce you to the latest version of the LEWICE ice accretion simulation software. We have worked long and hard to produce an accurate and reliable computational tool to help you with your aircraft icing analysis tasks. Additionally, the Icing Branch has and continues to work on development of an array of computational tools to assist in the analysis of aircraft encounters with icing conditions. We hope that the presentations and demonstrations at this workshop will provide you with a comprehensive look at our products and will help you to solve whatever icing simulation problems that you may encounter.

The workshop also allows you to let us know what your needs are in the field of computational simulation of aircraft icing. So please take the time to make your views known either through interaction with the presenters during the course of the workshop or during any of the breaks. We want to know what your needs are in order to better structure our research activities.

Thank you for attending and enjoy the workshop.

Outline

- Goals for the Workshop
- Software Development Activities
  - Ice Accretion Simulation
  - Ice Protection System Simulation
  - Effects of Ice Accretion on Aircraft Performance
- Validation Issues
- Concluding Remarks

This introduction is being provided to let you know what we will be covering over the course of the next two days.

First, we have set out some goals for the workshop with respect to what we would like to accomplish and what we hope will be provided to you during these two days. I'll mention those goals briefly and I hope that you can let us know at the end of the workshop if we have accomplished what we set out to do.

Next, I'll briefly describe the various computational simulation activities the Icing Branch is engaged in, whether via in-house activities or through cooperative relationships with grantees and contractors.

Then, I'll discuss our perspective on validation issues as they relate to computational simulation. Hopefully, you'll see through that discussion and through the work being presented over these next two days that we consider validation to be a critical element of the software develoment process.

Finally, I'll have a few concluding remarks summarizing our current computational simulation efforts.

Goals of the Workshop

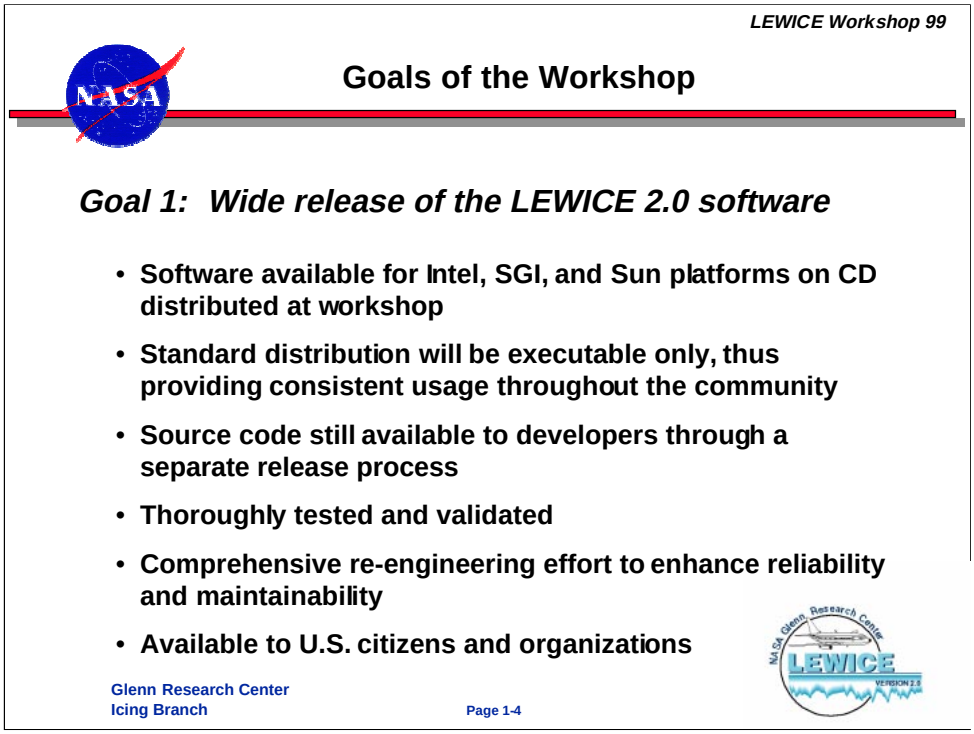
Goal 1: **Wide release of the LEWICE 2.0 software**

Goal 2: **Provide instruction in use of LEWICE 2.0**

Goal 3: **Increase awareness of complete line of Icing Branch computational tools**

Goal 4: **Provide information on Icing Branch computational simulation plans**

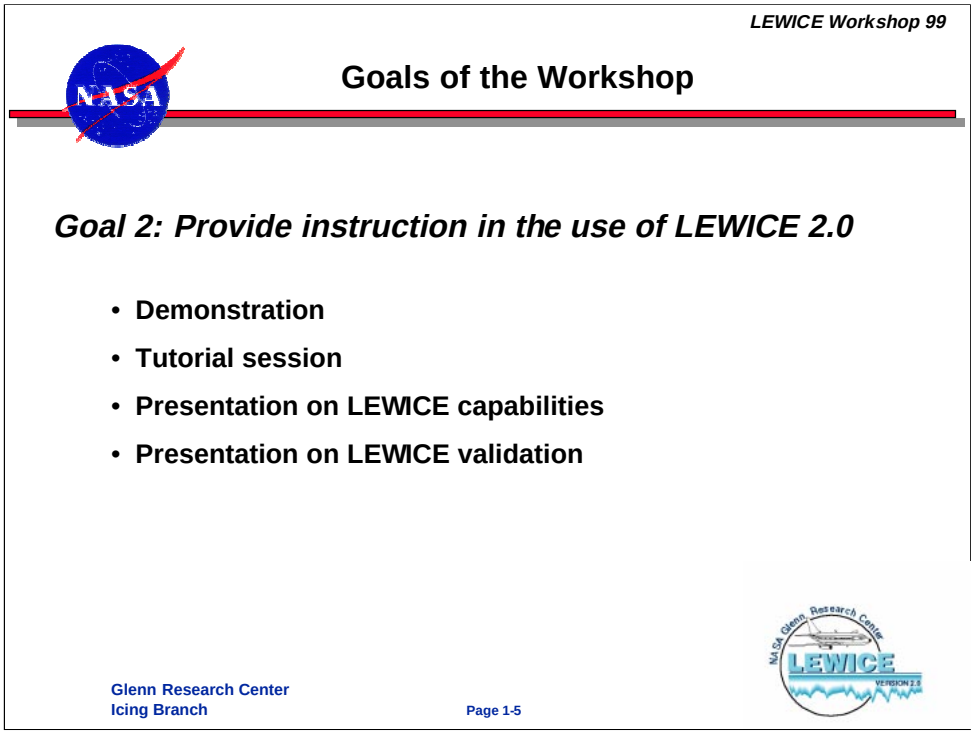
The four goals for the LEWICE Workshop are shown above. I will describe each of them individually and how we are planning on satisfying them in the next few slides. Please keep themin mind throughout the workshop and let us know well we did in satisfying them. We would like to know what we did right and what could be improved for the next time we have a workshop.



Our first goal is to have a general release of the LEWICE 2.0 software. The code has been released to some beta testers since January, 1999. Now we would like to distribute the official release of LEWICE 2.0 to all U.S. citizens and organizations.

From the text of the slide you can see that LEWICE is available on several platforms. Most people will be receiving the executable only. This is being done to address some of the concerns of the icing community with respect to version control and tracking that surfaced following the release of LEWICE 1.6. We will still provide source code to those individuals and organizations that need it, however, once the source code is out of our control we cannot provide the same level of support as we will for the official version of the code.

We are certain that this new release of LEWICE is the most well documented and well validated version of the code to date. We also feel confident that it is the most comprehensively tested icing simulation software available anywhere. Does that mean it is completely accurate and perfectly error free? No, of course not, no software is. However, it has been tested over the most extensive range of icing conditions and airfoil shapes available and we are certain that it will perform over that range of conditions and geometry types within the accuracy parameters that we have stated and which will be described for you later in the workshop.

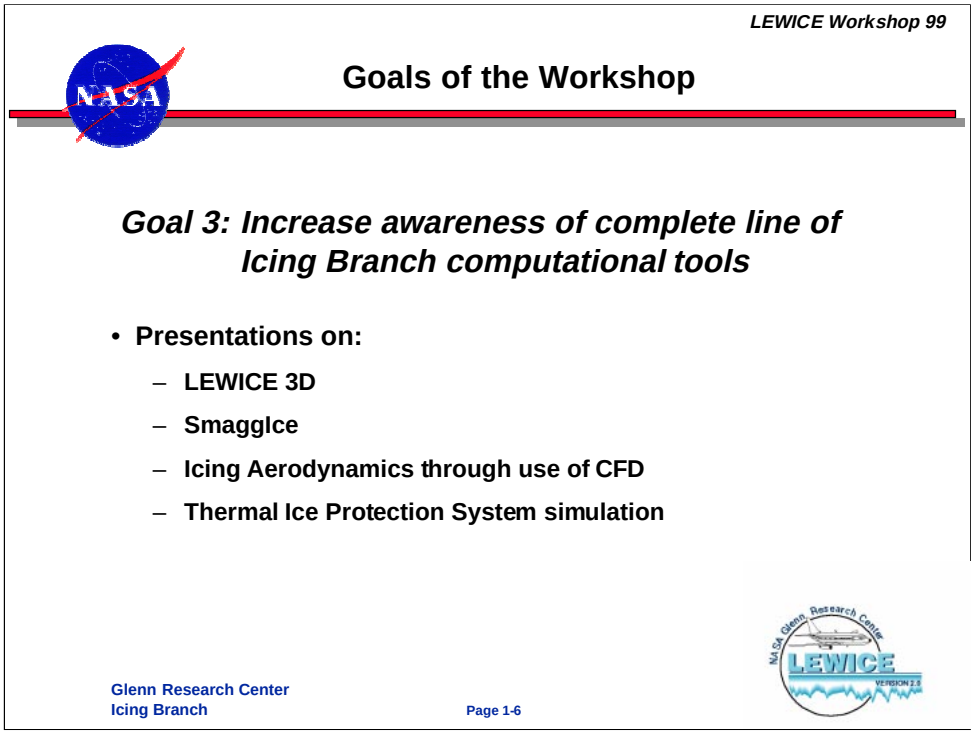


The workshop is designed to give a comprehensive User's perspective of the software. As such, we will be providing instruction in use of the software through the means described in the slide.

Demonstration of the software and how to set it up for operation will be provided through the general LEWICE presentation and in a training session held in the auditorium afer lunch.

A second training session, concluding the first day of the workshop, will provide hands-on experience in use of the code. We have PC's available for workshop participants to run the code and create ice shapes for some of the sample cases on the CD. There are enough PC's such that there should be only two or three people at any one PC.

Finally, the presentations on the LEWICE software are focussed on use of the code and the validation effort that was performed to increase the accuracy and reliability of the software. We will not be discussing, in detail, the science behind the software but rather will focus on how to use the software and what to expect from it.



We are also taking this opportunity to make you aware of the other computational simulation development efforts currently underway here at the NASA Glenn Research Center. Most of you are familiar with LEWICE but we have other tools available for analysis of different aspects of an icing encounter.

3D icing simulation tools: This consists our 3D ice accretion software LEWICE3D, a 3D Cartesian grid droplet trajectory code called ICEGRID, some plotting utilities called pltlew3d and lewplt3d. A new piece of software for 3D ice accretion simulation is currently under development called XXXXXX.

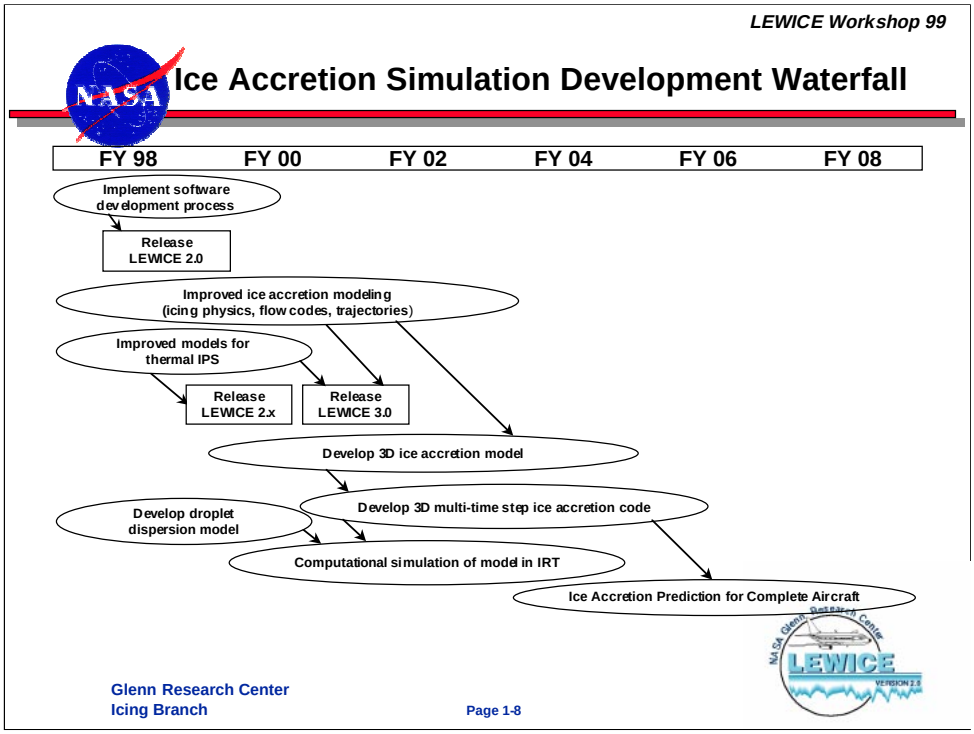
Grid generation and pre-conditioners: The use of grid based CFD tools for ice shape geometries has always been made difficult by the process of grid generation. SmaggIce is a tool currently under development at NASA Glenn that will assist in grid generation by pre-conditioning the coordinates of the geometry in order to allow creation of a well-formed grid without degrading the consequent aerodynamic analysis.

CFD analysis of icing aerodynamics: We have been using publicly available CFD tools to evaluate the aerodynamics associated with ice shape geometries. These tools are valuable assets in understanding the consequences of ice deposition on aircraft aerodynamics.

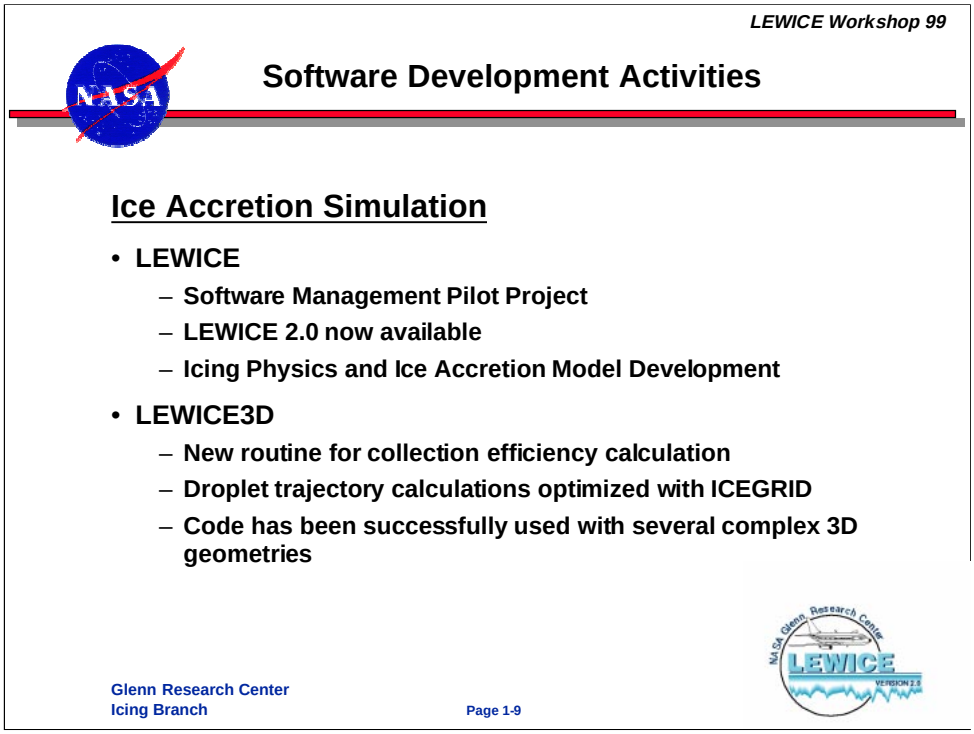
Thermal ice protection system simulation: The Icing Branch has two tools available for thermal IPS analysis. These tools are planned to become integrated into LEWICE as foolow-on efforts to the release of LEWICE 2.0.

We also hope to give you information related to our computational simulation development plans. This will be accomplished by the means listed in the slide. We want to insure that communications occur in both directions in order to better tailor our efforts to the needs of the icing community.

We would like to encourage you to take the opportunity to meet with the Icing Branch personnel while at the workshop and to continue that interaction when you return home.



This slide gives you a quick overview of our proposed development efforts over the coming years. We continue to work towards the goal of providing a complete simulation of an aircraft icing encounter. The timeline and ovals show the nature of our development activities and an indication of the anticipated length of each development activity. As you can see, with the current release of LEWICE 2.0 we have met one of our goals. Also, you can see that we will be working towards a version of LEWICE which will incorporate improved ice accretion models while at the same time we will be improving LEWICE 2.0 by adding thermal ice protection system components.



The most prominent element of our software development activities is our work on ice accretion simulation. These tools are used to determine ice growth on aircraft surfaces given the input geometry and information on the atmospheric conditions in the icing cloud. Both of the software packages mentioned above are based on the Messinger model of the ice growth process with some modifications resulting from icing physics experiments conducted in-house at NASA or by grantees and contractors.

LEWICE: The LEWICE software has been and continues to be the focus of much of the work here at NASA. The code has been re-engineered to conform to more stringent software design standards and has been thoroughly tested in order to enhance accuracy and reliability. LEWICE has been and continues to be the subject of a software management pilot project. By adopting well-established software management principles, the Icing Branch expects to enhance the maintainability and support for LEWICE through the use of appropriate documentation standards, version control practices, and release policies.

LEWICE3D: The 3D version of the LEWICE software has been developed as a separate simulation tool. This is mainly due to the differences in requirements on computation time, memory size, computational modeling methods, and appropriately sized computational hardware. As such, LEWICE3D is currently used in conjunction with an existing flow solution from a CFD code. LEWICE3D has several additional tools associated with it for droplet trajectory calculations and plotting. All of these tools have been used sucessfully on several cooperative programs between NASA and leading aerospace companies.

Software Development Activities

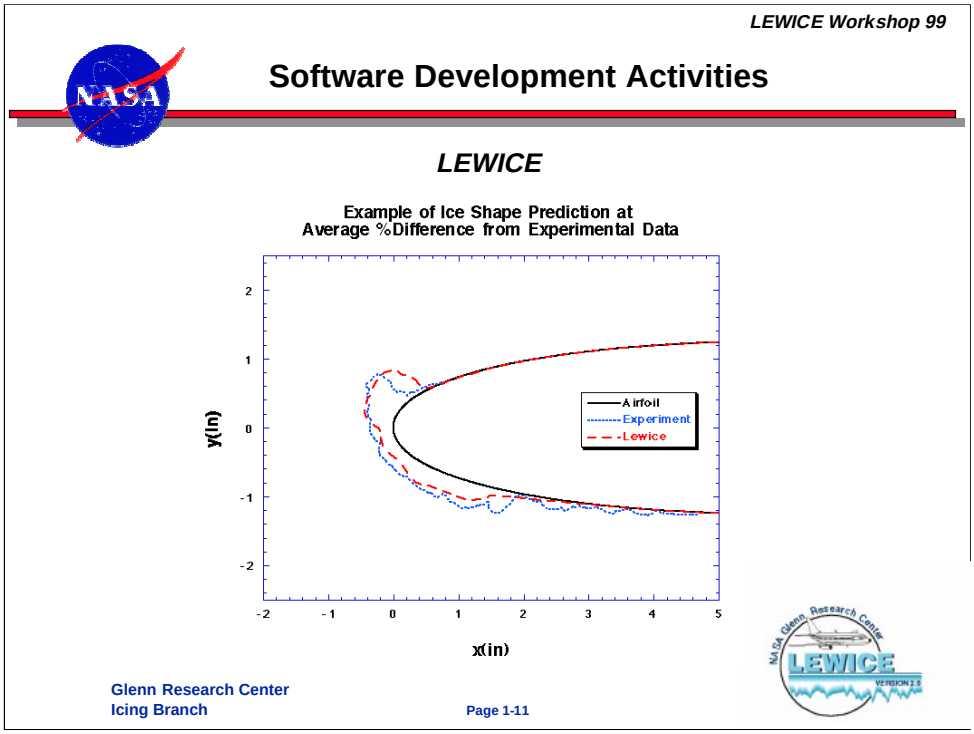
Ice Accretion Simulation

- **Numerical IRT (UIUC - Loth and Hancir)**
  - **Initial modeling of IRT aerodynamics and cloud characteristics has been completed**
  - **Effects of tunnel turbulence examined**
  - **Results of cloud uniformity comparisons are available**

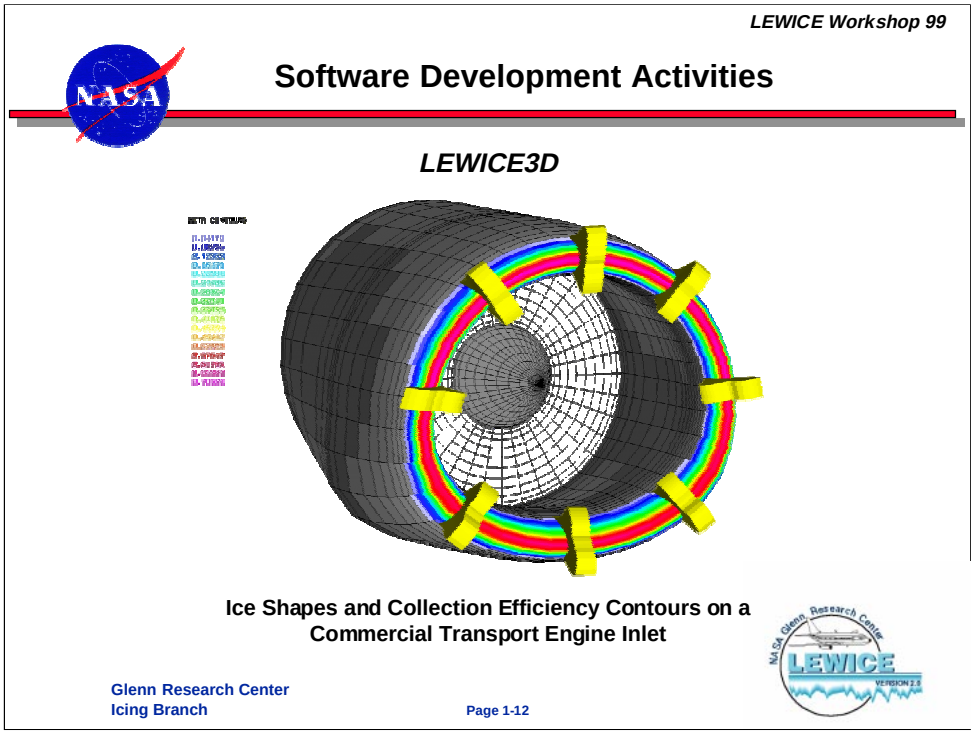
At an earlier stage of development, is our work on simulation of ice accretion within the test section of the NASA Icing Research Tunnel. This effort is being performed through a grant activity with Dr. Eric Loth of the University of Illinois. This work is directed at allowing simulation of the actual icing cloud in the IRT as well as test articles to be placed in the test section. By developing such a tool, analysts will be able to:

- Evaluate the effect of changes in the configuration of the tunnel on the icing cloud prior to implementation and;
- Examine icing patterns on test articles before actual testing commences
- Determine the most critical icing conditions to be examined in a test program during the test design phase

The current status of this project is described in the slide.



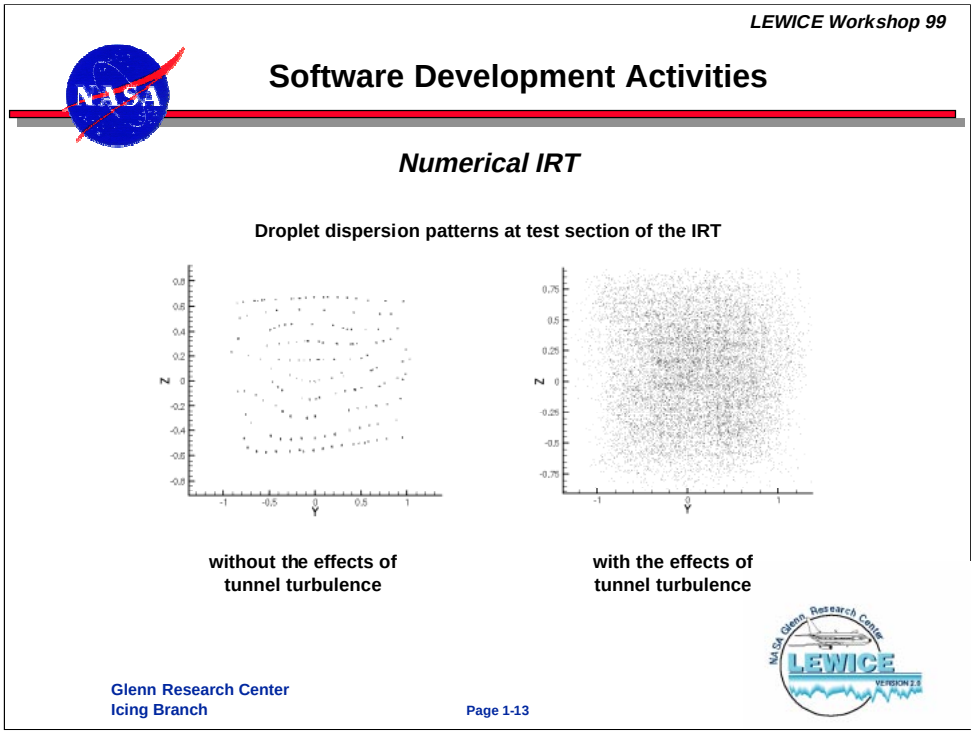
This slide shows a typical result of the LEWICE 2.0 ice accretion simulation software. The LEWICE ice shape is compared to an experimental ice shape, which was obtained from a test run in the IRT. As indicated, this example shows a result which has the same quantitative difference, based on the ice thickness measurement comparison technique, as the average difference from all of the LEWICE/experiment comparisons. Further discussion of these typical results will be covered by Bill Wright later in the workshop.



This slide shows the capabilities of NASA's 3D ice accretion simulation tools. In this example, LEWICE3D was used to predict the ice shapes and collection efficiency contours on a commercial transport engine inlet. The results indicate the variation in both those quantities as a function of circumferential location on the inlet lip. The LEWICE3D code uses 3D CFD results (whether potential flow, Euler, or Navier-Stokes computations) and calculates 3D droplet trajectories while using a strip approach to ice growth calculations to produce the results shown above. Currently, this only allows a single time step calculation of the ice growth.

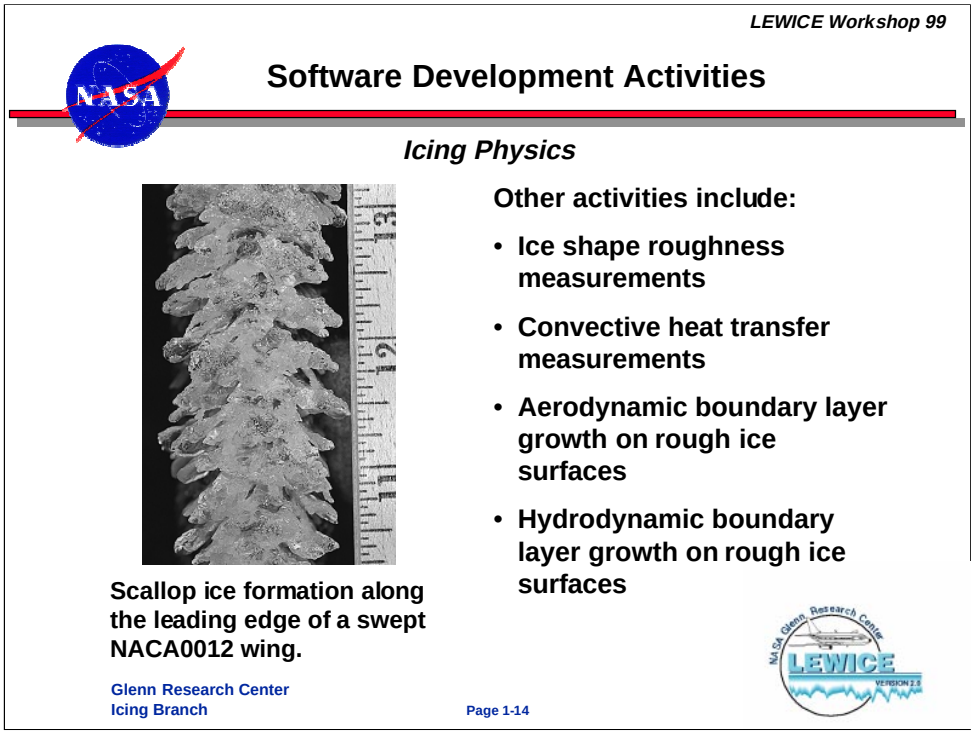
NASA has used these tools in cooperative activities with several major aerospace companies in order to examine the potential for icing on complex geometries such as engine inlets and radomes. Further information with respect to LEWICE3D and the other 3D icing tools will be provided later in the workshop by Colin Bidwell.

Further work is needed to construct a full 3D ice growth module and to incorporate that into a multiple time step calculation process.

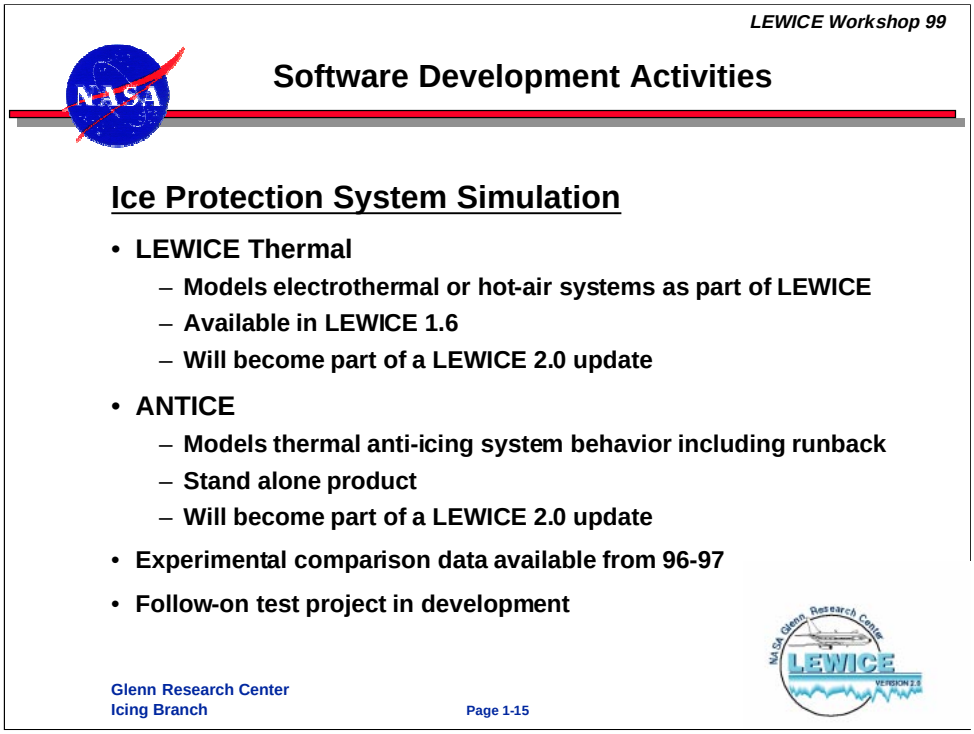


This slide shows the results of the simulation of the icing cloud in the IRT at a vertical plane in the center of the test section, both with and without the effects of tunnel turbulence. In this code, water droplets originate at the individual nozzles upstream of the test section between the heat exchanger and the contraction section. When the effects of tunnel turbulence are not included in the simulation, water droplets follow identical paths from a given nozzle to a specific location in the test section. The computations, shown on the left in the slide, indicate the unrealistic cloud distribution that would result. However, when tunnel turbulence is included, not all droplet paths from the nozzles are identical. This produces the more realistic pattern shown on the right in the slide.

Further work is required in order to examine the accuracy of the computational cloud pattern as well as to include the ability to model a test article in the overall simulation.



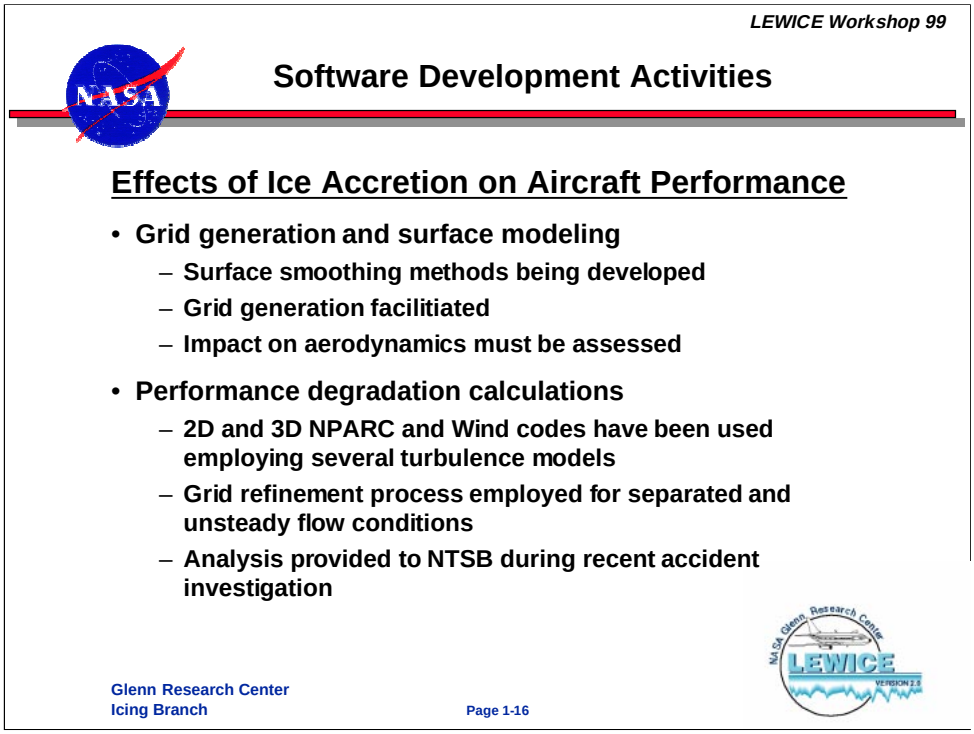
In order to develop updated computational models of the ice growth process, experimental activities related to the physics of icing are conducted in-house as well as through grant efforts. The physics of 2D rime ice growth are fairly well understood and models of that process appear to do an adequate job of predicting such accretions. However, the physics of glaze ice growth and 3D ice accretions are not quite as well known. As a result, experiments designed to quantify the effects of such characteristics as ice roughness and boundary layer development have been conducted. Although those experiments will not be covered in this workshop, it is important to understand that such work is ongoing and should find its way into the ice accretion simulations of the future.



The Icing Branch also has two thermal ice protection simulation tools available, as indicated in the slide. These tools were developed in the recent past as additions to the LEWICE software. They have not been included in the LEWICE 2.0 release because they have not undergone the re-engineering and testing processes applied to LEWICE. As indicated in the slide, our plans are to include them in an update to LEWICE 2.0 in the near future.

LEWICE Thermal is a 2D code that can be used to model de-icing or anti-icing behavior of electrothermal or hot air systems. It models multiple layers of material representing the thermal path between the inner surface and outer skin of an airfoil or other aircraft surface. It currently is included as a module within LEWICE 1.6 and will become part of LEWICE 2.0 following re-engineering and testing.

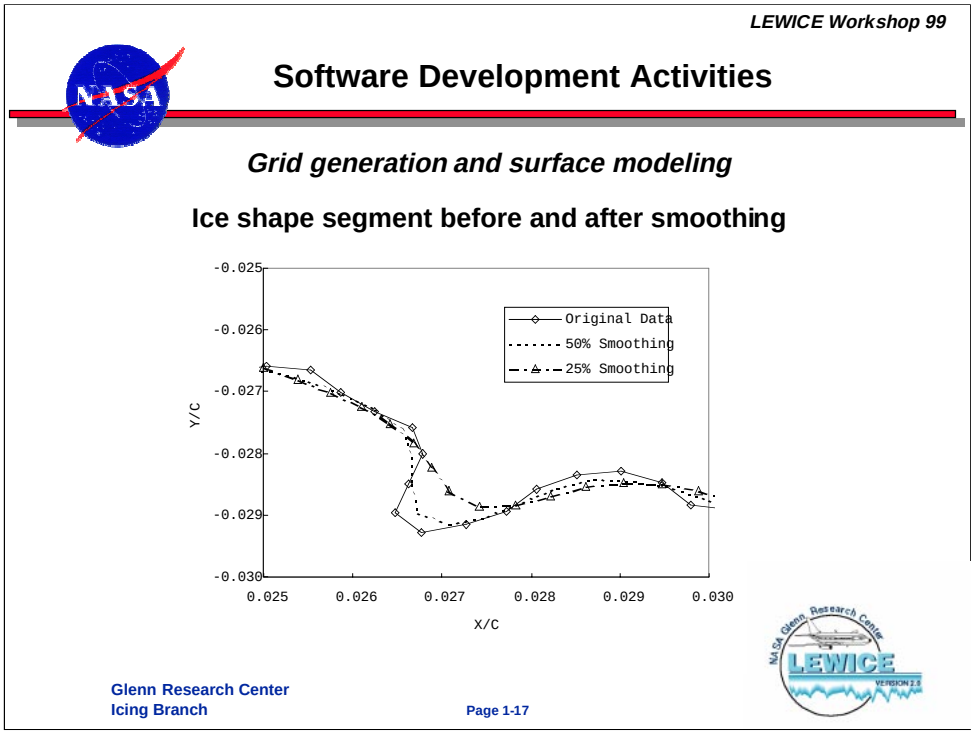
ANTICE is a 2D code that can be used to model anti-icing behavior of a thermal ice protection system which includes the effects of water running back on the surface. This differs from the LEWICE Thermal approach in that it includes the effects of rivulet formation. ANTICE is currently a stand alone code that can be used in conjunction with LEWICE. It is planned to include ANTICE as a module in a future release of LEWICE.



Once an ice shape has been determined, whether through computational simulation or from experiment, it is desirable to be able to examine the effects of that ice accretion on the aerodynamic performance of the body of interest. In all but the most benign conditions, a Navier-Stokes solver is needed to model the complex aerodynamics associated with ice shape geometries. Even with such tools available, the grid generation and turbulence modeling capabilities necessary for ice growh shapes go beyond what is available in the CFD community at large.

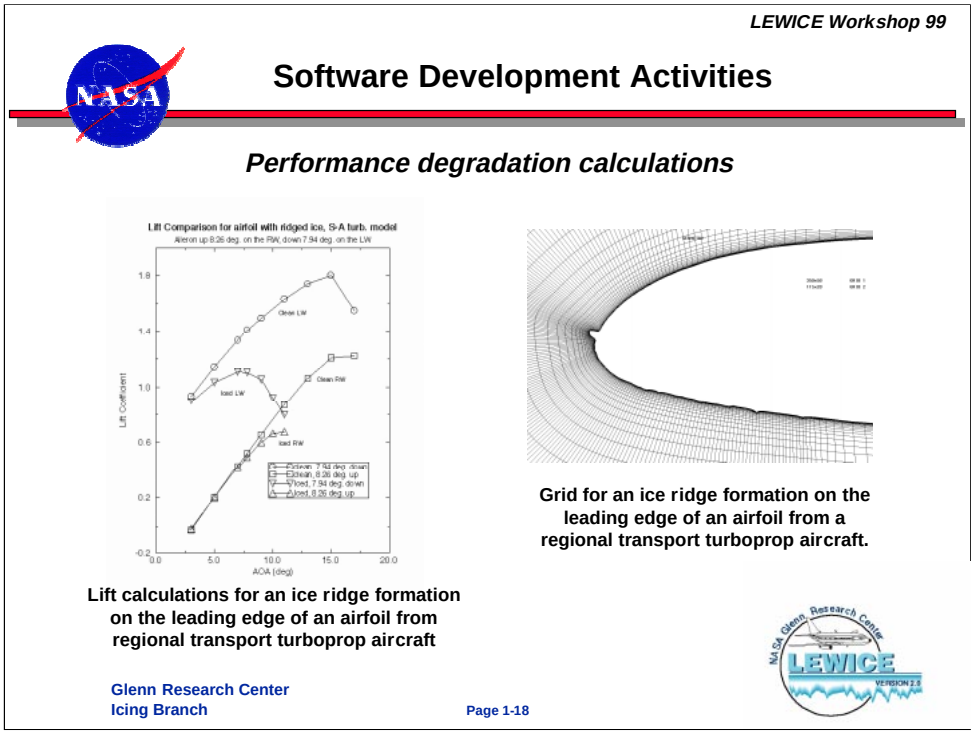
Grid generation is a particularly difficult process for ice shape geometries. The use of structured grid codes is desirable due to their relative efficiency and accuracy. However, this can be problematic when trying to construct a computationally “good” grid around the complex geometry of an ice shape. The Icing Branch is developing a tool that will allow the use of a structured grid by pre-processing the ice shape geometry in a way that does not substantially alter the shape yet will be more amenable to the grid generation process. Currently, that tool is interactive in nature with the eventual goal of automatic operation.

We have re-invigorated our efforts in CFD analysis of ice airfoil performance degradation. Work has centered on use of CFD tools to examine the behavior of wings and airfoils with ice shapes on the leading edge. Some recent activities have included an examination of the effects of geometry smoothing on aerodynamic characteristics and analysis in support of an accident investigation.



This slide shows the use of the SmaggIce code in pre-conditioning an ice shape geometry prior to grid generation. This is a highly magnified view of an ice roughness element that is part of a larger ice shape. The original data can result in a poorly formed grid due to the rapid changes in direction and slope surrounding the roughness element. Shown here are two levels of smoothing applied to that shape in order to provide a geometry that is more conducive to grid generation. The 50% smoothing level, shown in the slide, was eventually chosen due to an evaluation of the resulting grid and a determination that the effects on calculated aerodynamic properties were within the error expected from the solution procedure itself.

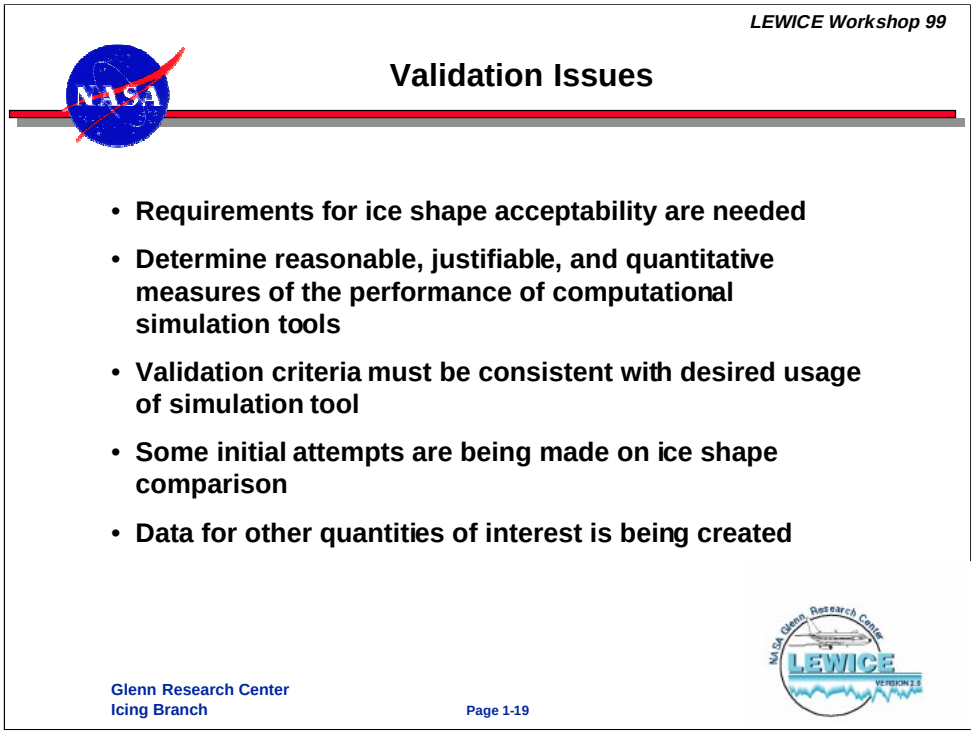
A more detailed description of the SmaggIce program will be provided later in the workshop. The Icing Branch contact regarding SmaggIce is Yung Choo.



CFD analysis of iced aircraft geometries has the potential to provide the information regarding the impact of the ice accretion on the aerodynamics of the vehicle. Once the grid generation process has been completed, the CFD analysis is performed to find out information on pressure distributions, lift, drag, pitching moment, etc. As shown in this slide, CFD analysis of the clean and iced airfoil identified the very large impact of a small ice ridge on the lift coefficient of a turboprop airfoil section.

Analysis such as this can help in assessment of candidate airfoil designs, in examination of an airfoil prior to actual testing in a wind tunnel, and in evaluation of airfoils after exposure to an icing incident.

More details will be provided later in the workshop. This work is being done by J.K. Chung.



The issue of validation of simulation methods will always play a role in development of investment strategies related to creation of computational tools. As such, the most difficult element of the validation process is knowing when you are done. In order to do that, it is necessary to understand what requirements must be satisfied in order to accept the software for use in a particular manner. The development of requirements and the acceptance criteria for those requirements is an ongoing and interactive process involving the software developer and the users of the software. Some of the qualities that play a role in the development are listed in the slide.

Validation Issues

Ice Accretion Simulation Validation

- **Modern Airfoil Program database available for ice shape comparison**
- **LEWICE 2.0 ice shape results compared with an extensive database of experimental ice shapes**
- **Collection efficiency database developed and analysis is underway**
- **Icing physics studies providing information for ice accretion model improvement**
- **Methods for assessment of ice shape comparison being developed**

The validation of our ice accretion simulation tools has been a high priority over the past 18 months and will continue to play a primary role in all future releases of the LEWICE software. Some of the activities associated with that effort are listed in the slide. Bill Wright will have a detailed discussion of the LEWICE validation testing and its results later in the workshop.

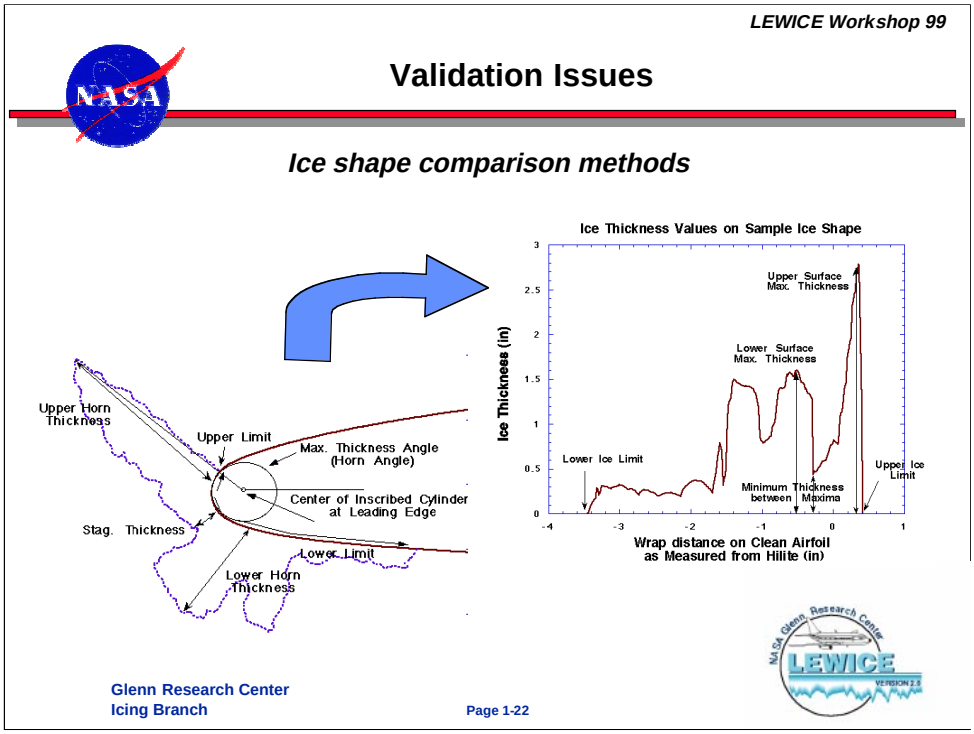
Validation Issues

Ice shape comparison methods

- **Quantitative comparisons employed for LEWICE 2.0 evaluation**
- **Alternate methods being examined by Ruff at Drexel**
- **Workshop held during April of this year; results suggest use of both feature comparison approach and of Ruff's FFT method**

One of the first items considered when evaluating ice accretion simulation software is how well do the ice shapes compare to measured shapes from experiment. Until recently this was always done in a qualitative manner by plotting the ice shape profiles on the same graph and making some visual assessment of the comparison. A workshop was held this past spring, at which three separate methods of comparing ice shapes based on measured quantities from each shape were presented. The outcome was that some quantitative methods were shown to be useful in making comparisons between ice shape profiles.

Application of these methods will eventually allow creation of some quantitative measure for acceptability of candidate ice accretion simulation methods.



This slide shows application of the ice thickness feature measurement method, developed by Bill Wright, to a sample ice shape profile from an IRT measurement. By translating the ice shape into an ice thickness plot, as shown on the right, two or more ice shapes can be compared on the basis of several easily identified features. The slid shows some of those feaures, such as, upper surface ice thickness, upper ice limit, and minimum thickness between maxima. By recording and comparing the quantities and their locations for large numbers of icing conditions, statistical information can be developed to indicate the capabilities of the software. Some of these statistical values will be talked about later in the workshop by Bill Wright. As these statistics are accumulated, capabilites of simulation tools can be more objectively evaluated and standards for acceptance can begin to be formulated.

Concluding Remarks

- **NASA's computational simulation tools are moving from research to design and analysis**
- **Software management practices are being adopted**
- **Specification of validation criteria is critical and requires input from icing community**
- **Continued development of tools for simulation of ice accretion, ice protection system performance, and icing effects**
- **Icing physics studies are continuing; development of a new model based on these studies is planned**

In conclusion, we are working hard to produce new products for icing simulation as well as to improve our existing tools with respect to accuracy and reliability. By adopting standard software design and management practices we expect continual improvement of our products and enhanced support for those products we deliver to the icing community.

Contacts

- **Branch Chief - Thomas Bond**
- **Computational simulation strategic plans - Mark Potapczuk**
- **LEWICE 2D and thermal codes - William Wright**
- **LEWICE 3D and 3D droplet trajectories - Colin Bidwell**
- **SmaggIce and grid generation for icing - Yung Choo**
- **CFD analysis for iced aircraft aerodynamics - J.K. Chung**